

# Customer Segmentation Using Unsupervised Learning

Data Science Project 3 - DecodeLabs Industrial Training Kit  
*PCA Dimensionality Reduction, K-Means Clustering and Business Persona Translation*

## Abstract

This project applies unsupervised learning to segment customers in an unlabeled e-commerce dataset of 1,200 orders (1,189 unique customers). Behavioural features were engineered, the data was one-hot encoded into 34 columns and standardised, and Principal Component Analysis (PCA) reduced it to 3 dimensions. The optimal number of K-Means clusters was determined mathematically using the Elbow Method and Silhouette Scores, leading to a choice of  $K = 3$  that balances statistical separation with business usability. The three resulting clusters were translated into actionable personas - Big-Basket Deal Maximisers, Full-Price Loyalists and Cautious Deal-Seekers, each with distinct spending levels, coupon behaviour and recommended marketing actions.

## 1. Introduction and Objective

Unlike supervised learning, this project provides no target column. The task is to let a distance-based algorithm discover natural groupings in raw retail data and then translate those mathematical groupings into customer personas a business can act on. The specific requirements were to: (1) apply PCA to reduce 20+ columns to 2–3 dimensions, (2) prove the optimal number of K-Means clusters using the Elbow Method and Silhouette Score and (3) translate the clusters into business personas.

## 2. Dataset and Feature Engineering

The dataset contains 1,200 e-commerce orders. Since 1,189 customers are unique, each order effectively represents one customer's purchase profile, so clustering orders is equivalent to clustering customers.

Beyond the original columns, behavioural features were engineered: UsedCoupon, CartConversion and Month and DayOfWeek extracted from the order date. Identifier columns were dropped and the categorical variables (Product, PaymentMethod, OrderStatus, CouponCode, ReferralSource) were one-hot encoded, producing 34 numeric columns. Because K-Means relies on Euclidean distance, all features were standardised with StandardScaler so that large-scale variables such as TotalPrice could not dominate binary features.

## 3. Methodology

### 3.1 Dimensionality reduction with PCA

PCA compressed the 34 standardised columns into 3 principal components. Examining the component loadings gave the axes clear meanings: PC1 is dominated by coupon-related features and separates coupon users from non-users (a “deal-seeking” axis), while PC2 is driven by TotalPrice, Quantity and ItemsInCart (a “spending size” axis). The three components retain about 18% of total variance - modest, which is expected for data with many near-independent one-hot columns and noted as a limitation.

### 3.2 Proving the optimal K

K-Means was run for K = 2 through 8 on the PCA-reduced data, recording inertia for the Elbow Method and the Silhouette Score for each K:

| Number of clusters K | Inertia | Silhouette Score |
|----------------------|---------|------------------|
| 2                    | 4,672   | 0.438            |
| 3                    | 3,196   | 0.380            |
| 4                    | 2,582   | 0.364            |
| 5                    | 2,099   | 0.362            |
| 6                    | 1,812   | 0.339            |

The silhouette score is highest at K = 2 (0.438), with K = 3 close behind (0.380) and the elbow in the inertia curve appears around K = 3. K = 2 would merely split coupon users from non-users - mathematically clean but too crude for marketing action. K = 3 was therefore selected: it retains strong separation while producing segments that differ meaningfully in both spending and coupon behaviour. This combination of statistical evidence and business reasoning is the required "proof" of optimal K.

## 4. Results: Cluster Profiles and Personas

The final K-Means model (K = 3) produced three well-differentiated segments. Average behaviour per cluster: Cluster 0 spends = 1,732 per order with 4+ items and 100% coupon usage. Cluster 1 spends = 1,043 with zero coupon usage. Cluster 2 spends = 488 with small quantities and full coupon usage.

| Cluster                | Persona                    | Behaviour                                                               | Recommended action                                                                            |
|------------------------|----------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 0 (408 customers, 34%) | Big-Basket Deal Maximisers | Highest average spend (=1,732), 4+ items per order, always use a coupon | Shift from blanket discounts to loyalty points - they buy big regardless                      |
| 1 (309 customers, 26%) | Full-Price Loyalists       | Solid mid-level spend (=1,043), never use coupons                       | Most profitable per sale; nurture with early access and premium service, not discounts        |
| 2 (483 customers, 40%) | Cautious Deal-Seekers      | Smallest orders (=488), low quantities, only buy with a coupon          | Targeted bundles and spend thresholds (e.g. free shipping over a minimum) to grow basket size |

A key strategic insight: the largest segment (Cautious Deal-Seekers, 40%) generates the least revenue per order, while the smallest (Full-Price Loyalists, 26%) pays full price every time. Treating all customers identically - for example, sending everyone the same discount - would simultaneously waste margin on Cluster 0, be ignored by Cluster 1 and under-serve the growth potential of Cluster 2.

## 5. Limitations

- The three principal components capture only ~18% of total variance, so much fine-grained structure is lost, clusters are moderately separated (silhouette < 0.5) rather than sharply distinct.
- With essentially one order per customer, loyalty over time cannot be measured. A production segmentation would use RFM features (Recency, Frequency, Monetary) built from repeat-purchase history.
- K-Means assumes roughly spherical clusters of similar size, algorithms such as DBSCAN or hierarchical clustering could be compared in future work.

## 6. Conclusion

This project delivered a complete unsupervised learning pipeline meeting all stated requirements: 34 encoded columns reduced to 3 principal components with interpretable meanings, an optimal cluster count proven with the Elbow Method and Silhouette Scores and three K-Means clusters translated into named business personas with concrete marketing recommendations. The central lesson of unsupervised learning is demonstrated end to end: the algorithm finds the mathematical structure, but the value comes from translating that structure into decisions a business can act on.